6D HOLOMORPHIC CHERN–SIMONS ON $\mathbb{C}^3$ BOUNDARY: $W_{1+\infty}$, $Y_{1,1,1}$, AND TOROIDAL DIM

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ABSTRACT. SCAFFOLD STUB. A codim-2 defect $C \subset \mathbb{C}^3$ in 6d holomorphic Chern–Simons theory induces a boundary chiral algebra isomorphic to $W_{1+\infty}$ at $\Psi = -\sigma_2$, central charge $c = 1$ (Sugawara). The normal bundle $N_{C/Y} = \mathbb{C}^2$ supplies the two spectral parameters. The toroidal chiral quantum group arises on the formal disc via Ding–Iohara–Miki (DIM) plus Schiffmann–Vasserot CoHA; the global $\mathbb{P}^1 \times \mathbb{P}^1$ lift is conditional on class-M chain-level holography (see `conj:uch-gravity-chain-level` in Vol II). Miki's S_3 symmetry is the Weyl group of the Ω-background.

PRIMARY THEOREMS

- 6d hCS codim-2 defect on $C \subset \mathbb{C}^3$ yields $W_{1+\infty}$ boundary algebra at $\Psi = -\sigma_2$: (48 engine tests + 1 HZ-IV-decorated independent-verification test; PR S_3 triality + ASV Newton- identity paths upgraded to genuinely numerical 2026-04-18).
- Toroidal chiral QG on formal disc: (via DIM + SV CoHA).
- Toroidal chiral QG on $\mathbb{P}^1 \times \mathbb{P}^1$: (conditional on class-M chain-level holography).
- $Y_{1,1,1}$ central charge $c = 0$, $\kappa_{\mathrm{ch}} = \Psi$: (correcting an earlier $c = 3$ assertion).
- Miura coefficient universality $(\Psi - 1)/\Psi$ at all spins $s \geq 2$: (Vol I `thm:miura-cross-universality`).

CHAPTER INPUTS (POINTERS)

```
\input{../chapters/examples/toric_cy3_coha.tex}
\input{../chapters/examples/toroidal_elliptic.tex}
\input{../chapters/examples/k3_quantum_toroidal_chapter.tex}
\input{../chapters/theory/braided_factorization.tex}
```

CROSS-VOLUME DEPENDENCIES

- Vol I standalone `cy_quantum_groups_6d_hcs.tex` (programme viewpoint; Vol III extends with CoHA implementation).
- Vol I `thm:miura-cross-universality` (Miura coefficient universality).
- Vol II `\ref{part:ht-landscape}` (6d hCS as 5d HT shadow).

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